NATIONAL JUNIOR COLLEGE
SENIOR HIGH 2 PRELIMINARY EXAMINATION
Higher 2

NAME

SUBJECT CLASS: 2ma2

REGISTRATION NUMBER

MATHEMATICS 9758/02

Paper 2 **14 September 2026**

3 hours

Candidates answer on the Printed Answer Booklet.

Additional Materials: Printed Answer Booklet
List of Formulae and Results (MF27)

READ THESE INSTRUCTIONS FIRST

Write your name, class and registration number on the work you hand in.
Write in dark blue or black pen.

Answer **all** the questions.
Write your answers on the Printed Answer Booklet. Follow the instructions on the front cover of the answer booklet.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use an approved graphing calculator.
Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphing calculator are **not** allowed in a question, you must present the mathematical steps using mathematical notations and not calculator commands.
You must show all necessary working clearly.

You are reminded of the need for clear presentation in your answers. Up to 2 marks may be deducted for improper presentation.

The number of marks is given in brackets [ ] at the end of each question or part question.

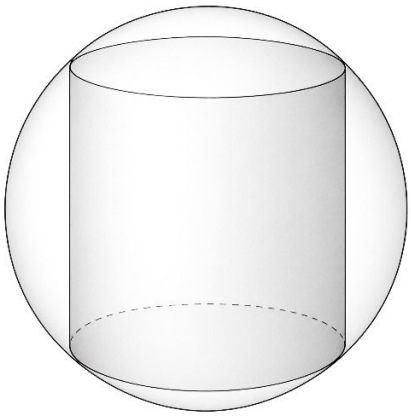
© NJC 2026 **[Turn over**

Section A: Pure Mathematics [40 marks]

1 A curve C has equation $ay^2 + bxy + cx^2 + d = 0$, where a, b, c and d are non-zero constants. The curve C passes through the point $\left(\frac{1}{\sqrt{2}}, \frac{5}{\sqrt{2}}\right)$, and the equation of the normal to C at its positive y-intercept is $x + y = \sqrt{10}$.

Find one possible set of integer values of a, b, c and d. [5]

2 A cylinder is inscribed in a sphere of fixed radius R. The cylinder has radius r and height h.



Find the exact maximum possible volume of the cylinder in terms of R. [6]

3 **(a)** It is given that $\sum_{r=1}^{n} \frac{1}{r(r+1)(r+2)} = \frac{n(n+3)}{4(n+1)(n+2)}$, where n is a positive integer.

(i) Prove that $(r+1)^3 > r(r+1)(r+2)$ for all positive integers r. Hence, show that $\sum_{r=1}^{n} \frac{1}{(r+1)^3} < \frac{1}{4}$ for any positive integer n. [3]

(ii) For $n \geq 5$, find $\sum_{k=5}^{n} \frac{1}{k(k^2-1)}$ in terms of n. [3]

(b) The Root Test states that a series of the form $\sum_{r=1}^{\infty} a_r$ converges when $\lim_{n\to\infty} \sqrt[n]{|a_n|} < 1$, and diverges when $\lim_{n\to\infty} \sqrt[n]{|a_n|} > 1$. When $\lim_{n\to\infty} \sqrt[n]{|a_n|} = 1$, the test is inconclusive. Using the test, determine whether the series $\sum_{r=1}^{\infty} \left(\frac{-3r}{5r+1}\right)^r$ converges or diverges. [3]

© NJC 2026

4 It is given that $\mathrm{f}(x) = \ln(1 - \sin 2x)$.

(a) Using standard series from the List of Formulae (MF27), find the Maclaurin series for $\mathrm{f}(x)$ up to and including the term in x^3. By referring to the Maclaurin series for $\mathrm{f}(x)$, explain whether the curve $y = \mathrm{f}(x)$ is concave upwards or downwards at $x = 0$. [4]

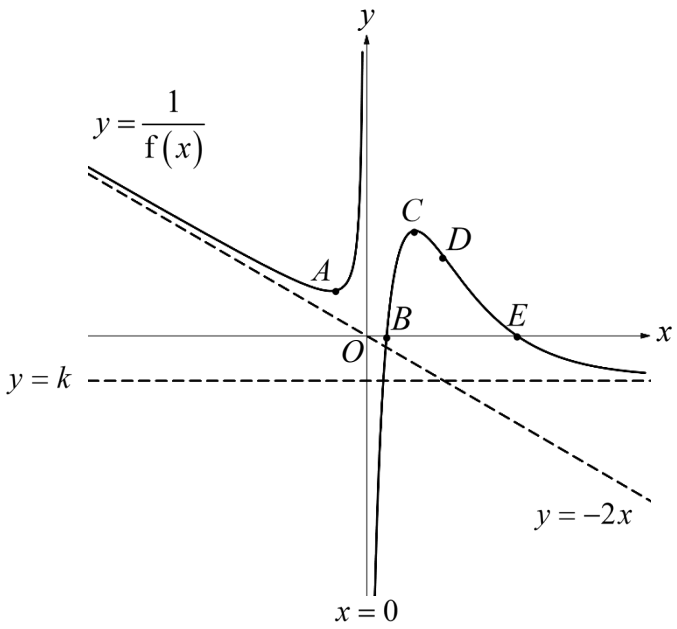
(b) It is given that the first three non-zero terms of the series expansion of $\mathrm{f}'(x)$ are equal to the first three non-zero terms of the series expansion of $\mathrm{e}^{ax^2}(b + cx)$, where a, b and c are constants. Find the values of a, b and c. [3]

(c) Hence, for sufficiently small values of k, show that

$$\int_0^k -\mathrm{e}^{ax^2}\left(bx + cx^2\right) \mathrm{d}x \approx \alpha k^2 + \beta k^3 + \gamma k^4$$

where α, β and γ are constants to be determined exactly. [3]

5 The diagram below shows the curve with equation $y = \dfrac{1}{\mathrm{f}(x)}$. The curve has turning points at $A(-2,5)$ and $C(3,8)$, intersects the x-axis at $B(1,0)$ and $E(9,0)$, and has a point of inflexion at $D(5,6)$. The gradient of the curve at D is -4. The equations of the asymptotes of the curve are $x = 0$, $y = -2x$ and $y = k$ where $k < 0$.



On separate diagrams, sketch the graphs of

(a) $y = \mathrm{f}(x)$, [4]

(b) $y = \dfrac{\mathrm{d}}{\mathrm{d}x}\left(\dfrac{1}{\mathrm{f}(x)}\right)$, [3]

labelling clearly the coordinates of the points corresponding to A, B, C, D and E where possible, and the equations of any asymptotes.

(c) Hence, find the value of $\mathrm{f}'(5)$. [3]

© NJC 2026

Section B: Probability and Statistics [60 marks]

6 Consider a fair 12-sided die with faces labelled 1, 2, 3, …, 12. The die is rolled once, and the number shown on the top face of the die is denoted by the random variable X.

(a) State $\mathrm{E}(X)$. [1]

(b) Given that $Y = 6(X-1)$, use the result $\sum_{r=1}^{N} r^2 = \frac{N}{6}(N+1)(2N+1)$ to find $\mathrm{Var}(Y)$. [4]

7 A computer game consists of three levels. A player always starts at Level 1. The probability that a player clears Level 1 is p. If a player clears a level, the probability of clearing the next level is halved. If a player fails a level, the game ends immediately.

A player who clears all three levels is awarded an "A" rank. A player who clears at least one level but did not obtain an "A" rank is awarded a "B" rank.

(a) The probability that a player who is awarded a "B" rank clears exactly two levels is $\frac{5}{17}$.

Find the value of p. [3]

(b) Using the value of p found in part **(a)**, find the probability that, out of 10 randomly chosen players, exactly 3 players were awarded an "A" rank. [3]

8 A total of 8 employees, including Celeste, Divya, Enqi and Farid, attend a meeting.

(a) The employees are seated in a circle. Find the number of different arrangements such that Celeste and Divya are seated next to each other, or Enqi and Farid are seated next to each other, but not both. [4]

(b) After the meeting, the 8 employees divide themselves into three groups, of which two groups have exactly three employees and one group has exactly two employees. Celeste and Divya must not be in the same group.

Find the number of different ways to form the groups. [3]

© NJC 2026

9 A property agent is investigating the relationship between the price per square metre of a terrace house and its distance from Central London. He collects information about 6 terrace houses, and the data is represented in the table below; the distance, x, is in miles and the price per square metre, y, is in thousand-pound sterling.

x	2.5	5	9	13	19	29
y	11.3	8.8	7.3	5.8	4.8	4.1

(a) Sketch a scatter diagram of the data. [1]

(b) Determine which of the following models is the better fit to the data where a, b, c and d are constants.

$$\text{Model (A): } y = \frac{a}{x} + b$$

$$\text{Model (B): } y = ce^{-\frac{x}{10}} + d$$

[2]

(c) Find the regression equation for the model identified in part **(b)**. [2]

(d) Use your equation from part **(c)** to estimate the price per square metre for a terrace house that is 32 miles from Central London, giving your answer correct to the nearest pound. Explain whether your estimate is reliable. [2]

(e) A new data point is added to the property agent's dataset. This new data point is for a terrace house 50 miles from Central London and yet has a significantly higher price per square metre compared to other terrace houses in the vicinity. Explain how the addition of this data point may affect the gradient of the regression line from part **(c)**. [2]

© NJC 2026

10 At the Central Manpower Base (CMPB) in Singapore, pre-enlistees undergo medical screening before National Service enlistment. Each day, a fixed number of randomly chosen pre-enlistees' blood samples are screened for medical conditions that influence vocational deployment, and the number of blood samples found to have an abnormal haemoglobin level is denoted by X.

(a) State, in context, two assumptions needed for X to be well modelled by a binomial distribution. [2]

Assume now that X has the distribution $\mathrm{B}(60, p)$.

(b) Given that the most probable number of blood samples with an abnormal haemoglobin level is 2, find exactly the possible range of values of p. [4]

In addition to haemoglobin levels, the same 60 blood samples are also screened for Glucose-6-phosphate dehydrogenase (G6PD) deficiency.

It is now given that for a randomly chosen blood sample, the probability that it has an abnormal haemoglobin level is 0.04 and the probability that it has G6PD deficiency is 0.05. It may be assumed that whether a blood sample has an abnormal haemoglobin level is independent of whether it has G6PD deficiency.

(c) Find the probability that, in a randomly chosen day, more than 4 blood samples exhibit at least one of the two medical conditions above. [4]

11 A global luxury company produces two signature blends, Baca coffee and TWJ tea. The mass, in grams, of a Baca coffee gift pack follows the normal distribution $\mathrm{N}(120, 5)$, and the mass, in grams, of a TWJ tea gift pack follows the normal distribution $\mathrm{N}(100, 3)$.

(a) Given that the probability that a randomly chosen Baca coffee gift pack weighs more than a grams is less than 0.9, find the range of values of a. [1]

(b) Find the probability that the total mass of two randomly chosen Baca coffee gift packs is within 47 g of thrice the mass of one randomly chosen TWJ tea gift pack. [3]

(c) State an assumption for the calculations in part **(b)** to be valid. [1]

The Baca coffee and TWJ tea gift packs are sold in tin boxes. The masses, in grams, of the empty tin boxes are normally distributed such that 95% of them have a mass less than 307.4 grams and 5% of them have a mass less than 292.6 grams.

(d) State the mean mass of the empty tin boxes and show that the standard deviation of the masses of the empty tin boxes is 4.50 grams correct to 2 decimal places. [3]

(e) A tin box is packed with m Baca coffee gift packs and n TWJ tea gift packs, for a total of 12 gift packs altogether. Find the values of m and n that maximises the probability that the total mass of the tin box and its contents lies between 1.58 kg and 1.62 kg. [3]

© NJC 2026

12 A technology company states that its new smartwatch can last an average of 500 minutes on a single charge when its Global Positioning System (GPS) mode is switched on continuously. A consumer association representative suspects that the stated mean battery life has been overstated.

The battery life of a smartwatch, X minutes, after a full charge with GPS mode switched on continuously, may be assumed to be normally distributed. A random sample of 50 smartwatches is tested, and the results are summarised as follows:

$$\sum(x-500)=-102 \qquad \sum(x-500)^2=3408$$

(a) Calculate unbiased estimates of the population mean and variance of the battery lives of the smartwatches. [2]

(b) Test, at the 6% level of significance, whether the consumer association representative's suspicion is correct. You should state your hypotheses and define any symbols that you use. [4]

(c) Explain, in the context of the question, the meaning of 'at the 6% level of significance'. [1]

A quality manager is told only that the test in part **(b)** rejected the null hypothesis at the 6% significance level. He does not know the test statistic or the sample data.

(d) Explain whether he can determine the conclusion of a test, at the same significance level, for whether the mean battery life differs from 500 minutes. [1]

After a software update, the battery life of the smartwatch is no longer normally distributed. A large random sample of n smartwatches is tested, and it is found that the battery life for this sample has mean 500.8 minutes and variance 9 minutes^2.

(e) A test at the 4% level of significance found sufficient evidence that the mean battery life of the smartwatches is greater than 500 minutes. Find the set of possible values of n. [4]

© NJC 2026

BLANK PAGE

© NJC 2026